

Object Detection Deployment Metrics

A practical checklist for benchmarking an object detection model before deploying it on edge hardware or a robot.

1. Model Size

Measure model file size (MB), number of parameters, and computational complexity such as FLOPs/MACs. These indicate storage requirements, memory requirements, and computational demand.

2. Detection Accuracy

Core detection metrics: mAP@0.5, mAP@0.5:0.95, precision, recall, and F1-score. For fruit detection, recall is especially important when missing an object is costly.

3. Inference Latency

Measure the time required for one inference. Also separate preprocessing, inference, and postprocessing latency when possible. Approximate model-only FPS can be estimated as $\text{FPS} = 1000 / \text{latency}(\text{ms})$, but end-to-end FPS should include the whole pipeline.

4. Memory Usage

Measure system RAM and, when applicable, GPU/accelerator memory. Record memory usage while the application is actually running inference.

5. Power Consumption

For battery-powered robots, measure voltage and current. Power can be estimated as $P = V \times I$. Power consumption directly affects battery life and thermal requirements.

6. Temperature and Thermal Stability

Run continuous inference for an extended period and record temperature and performance over time. Thermal throttling can reduce FPS after sustained operation.

7. End-to-End Latency

For robotics, measure the complete pipeline: camera capture → preprocessing → inference → postprocessing → coordinate calculation → communication/robot command.

$$T_{\text{total}} = T_{\text{capture}} + T_{\text{preprocess}} + T_{\text{inference}} + T_{\text{postprocess}} + T_{\text{communication}}$$

8. Robustness Testing

Do not test only clean validation images. Evaluate performance under different lighting, distances, occlusion levels, camera motion, backgrounds, and viewing angles.

9. Detection-Specific Analysis

When supported by the evaluation framework, measure AP for small, medium, and large objects. Also inspect missed detections, false positives, duplicate detections, and overlapping objects.

10. 3D Localization for a Future Robot

If depth sensing is added, measure physical localization error. A simple metric is $E = ||p_{\text{pred}} - p_{\text{true}}||$, where p is the 3D position.

Recommended Minimum Benchmark

Category	Metric
Accuracy	mAP@0.5
Accuracy	mAP@0.5:0.95
Accuracy	Precision
Accuracy	Recall
Accuracy	F1-score
Model	Parameters
Model	Model size
Compute	FLOPs / MACs
Speed	Preprocessing latency
Speed	Inference latency
Speed	Postprocessing latency
Speed	End-to-end latency
Speed	FPS
Memory	RAM usage
Memory	VRAM / accelerator memory
Hardware	CPU utilization
Hardware	GPU/NPU utilization
Hardware	Power consumption
Hardware	Temperature
Robustness	Lighting / distance / occlusion / motion

Priority for Your Current Apple Detector

If you do not yet have edge hardware, begin with: mAP@0.5, mAP@0.5:0.95, precision, recall, F1, model size, parameter count, inference latency, RAM usage, and FPS. After obtaining deployment hardware, add power, temperature, CPU/NPU utilization, and true end-to-end latency.